SMU Lyle School of Engineering

Reinforcement Learning

Course Intro

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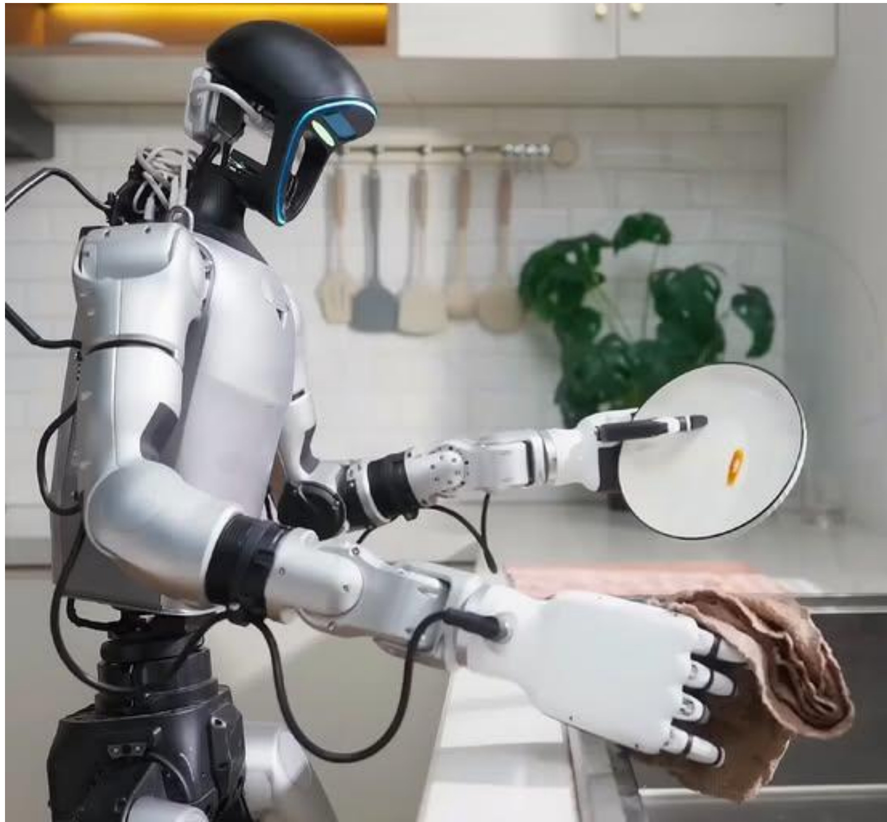
Course Goal

How Do We Build Intelligent Systems?

Options:

- Explicitly program behavior: **Rule-based agent, planning**
- Learn behavior from labeled data offline: **Supervised learning**
- Learn behavior using feedback from interacting with the environment (online): **Reinforcement Learning**



Course Learning Outcomes

- CLO 1: Explain foundational concepts of **reinforcement learning**
- CLO 2: Formulate sequential decision-making problems as **Markov Decision Processes** (MDPs)
- CLO 3: Apply **dynamic programming and Monte Carlo methods** to solve reinforcement learning problems
- CLO 4: Implement and evaluate **Temporal-Difference (TD) learning** algorithms
- CLO 5: Utilize **function approximation** methods for large or continuous state spaces
- CLO 6: Implement reinforcement learning agents with **eligibility traces and policy gradients**
- CLO 7: Critically evaluate current research and applications of **deep reinforcement learning**

Learning Method

Lecture + Problem Sets: We will discuss algorithms appropriate for different tasks and types of environments. Problem sets will improve the understanding.

Projects: You will implement algorithms to solve several tasks and conduct experiments to investigate how well the algorithms work and how they scale with problem size.

Exams: Checks knowledge of AI concepts and the ability to apply them.

Evaluation is based on problem sets, projects, and exams.

Course Mechanics

- **Office hours**:
  - My office hours can be found on my home page.
  - Weekly TA office hours will be done using Zoom. The time and the link will be on Canvas soon.
- **Communication**: Use Canvas messages and leave comments for assignments.

- **Assignment Grading**: The TA will grade assignments and problem sets. Please contact the TA (e.g., during office hours or by leaving a comment) to discuss your assignment and grade.

- **Final Grade**: The calculated grade on Canvas serves as a rough guide. I will assign your final grade at the end of the semester.

To Be Successful you need…

- Proficiency in **Python** programming.
- Familiarity with **artificial intelligence** fundamentals, including intelligent agents and search.
- Knowledge of **machine learning** (e.g., supervised learning, deep learning, model evaluation).
- Knowledge of **math and probability theory** (probability, expectation)